

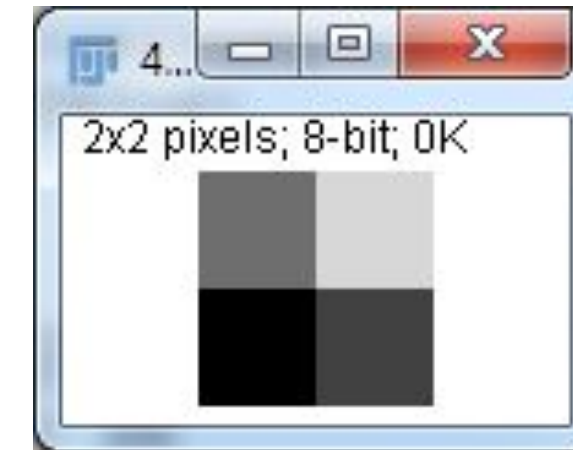
Image Resizing

Image Processing & Analysis for Life Scientists

Olivier Burri, Romain Guiet & Arne Seitz

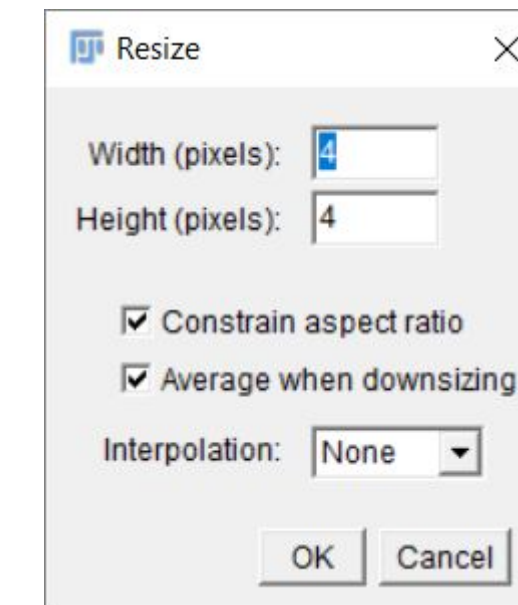
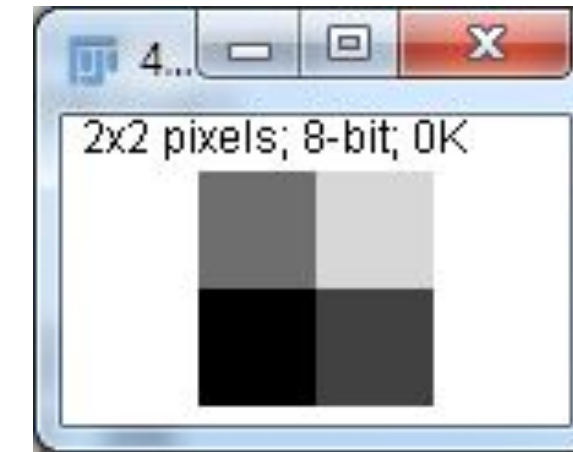
Upscaling

1. Open 4pixels.tif
2. Zoom in at maximum



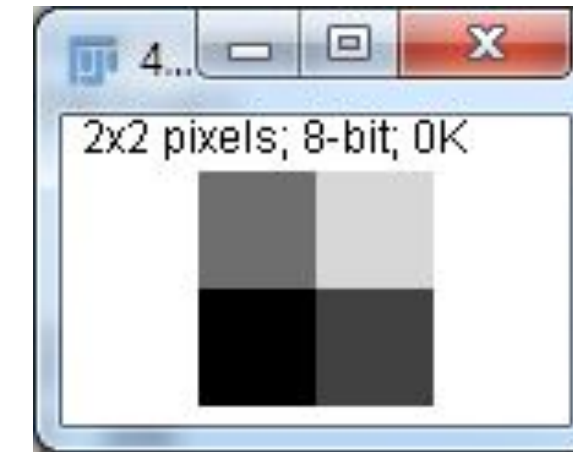
Upscaling

1. Open 4pixels.tif
2. Zoom in at maximum
3. Duplicate the image:
Image → Duplicate
4. Select : **Image → Adjust → Size**
 - Width: 4
 - Interpolation : None



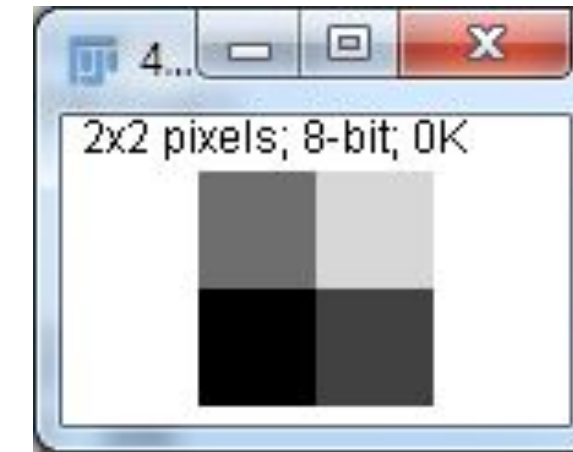
Upscaling

1. Open 4pixels.tif
2. Zoom in at maximum
3. Duplicate the image:
Image → Duplicate
4. Select : **Image → Adjust → Size**
 - Width: 4
 - Interpolation : None
5. Duplicate the original again
and adjust the size to a width of 3



Upscaling

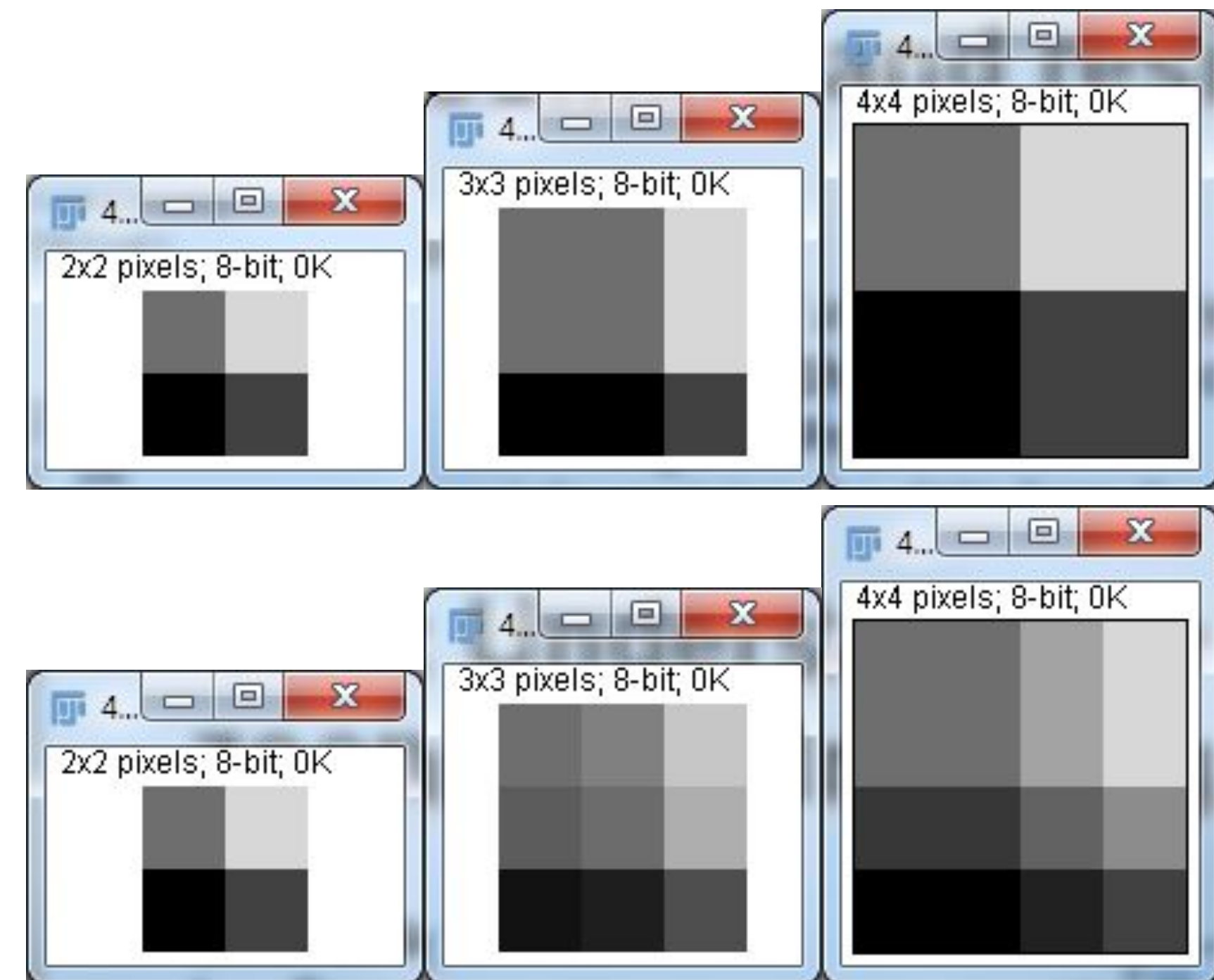
1. Open 4pixels.tif
2. Zoom in at maximum
3. Duplicate the image:
Image → Duplicate
4. Select : **Image → Adjust → Size**
 - Width: 4
 - Interpolation : None
5. Duplicate the original again
and adjust the size to a width of 3



Repeat 3-5 with interpolation.

Scaling

- Scaling modifies the values of the pixels of the image, so you should be careful when scaling up as much as when scaling down.
- Interpolation methods can alter scientific results depending on the method.



Down/Up Scaling

- Use : **Image** → **Scale...**
shortcut : Ctl+E
- Compare effects of :
 - Downscaling : scale between 0 and 1
 - Upscaling : scale above 1
- Look at the effects of :
 - Averaging
 - Interpolation

shapes.tif

